

Reliability Engineering 101 for NSWC Corona

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Part II : Reliability of Components and Systems

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Part I: Introduction and Mathematics

Module 1.1 : Scope of Reliability Engineering

Module 1.1.1 : NAVSEA Documentation on Reliability

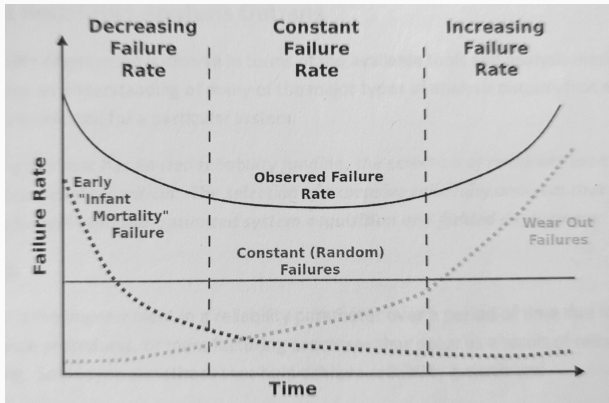
Reliability Engineering : *When you engage the ignition in your vehicle in the morning, how do you know it will work?*

Example documentation :

- “A Simplified Technical Guide for System Reliability: A Reliability Planning and Deployment Reference for Acquisition Professionals.” (Section 219 Project, 2019, Port Hueneme/NAWC China Lake/NSWC Corona)
- *System Reliability Toolkit-V. Quanterion Solutions Inc., 2015.*
- O’Conner and Kleyner : *Practical Reliability Engineering*. Wiley and Sons (2012).
- Modarres, Kaminskiy, and Krivstov : *Reliability Engineering and Risk Analysis, 3rd Ed.* CRC Press (2017).

Module 1.1.2 : Impact of Reliability

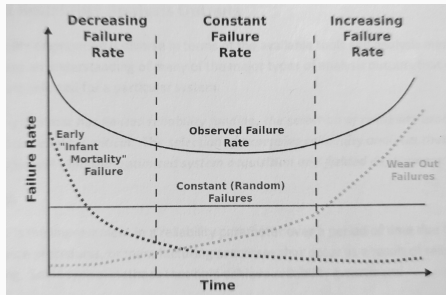
- Reliability drives efficient programmatic decisions.
 - Evidence-based acquisition activities
 - Optimization of design features and cost versus risk
 - Identification of design flaws and manufacturing defects
 - Enhancement of reliable system service-life



Module 1.1.2 : Impact of Reliability

Reliability engineering addresses different system types.

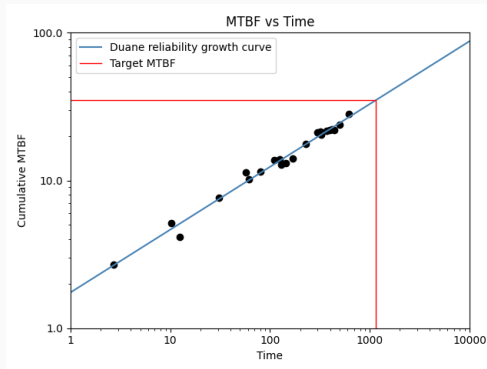
- *Impulse untestable systems* : must be irreversibly expended to measure their current state (operational or failed) to assess reliability.
- *Impulse testable systems* : the system must be tested to know its state (operational or failed), and testing can take place at sparse intervals
- *Continuous use systems* : the system state (operational or failed) is always known



Module 1.1.2 : Impact of Reliability

Reliability can grow by acting on analysis.

- **Reliability growth** : is an improvement in a reliability parameter over a period of time, driven by reliability growth programs (RGPs).
 - *Failure Modes Effects Analysis (FMEA/FMECA)* (criticality)
 - *Failure Report Analysis and Corrective Action System (FRACAS)*



Other common terminology in reliability :

- *Maintenance policy optimization* : analysis of maintenance procedures, including sparing, personnel, and transportation, to meet reliability requirement
- *Reliability predictions* : calculations and simulations of reliability parameters that lead to system insights
- *System availability* : quantifies how well systems are maintained such that they are ready to deploy. Depends on system type and is distinct from reliability measures.
- *Stockpile availability* : similar to system availability, but targets sub-groups of assets, rather than the entire set of assets.

System Types

- **Continuous Use** : These systems are maintained so that reliability parameters can be determined with continuous functions.
 - Time to failure or time between repair data
 - Potential failure data can be right-censored if it is unknown when the asset will fail in the future.
- **One-Shot/Impulse Untestable** : The reliability of these systems cannot be measured until the system is deployed irreversibly.
 - Failures are left-censored if it is unknown when the asset first failed.
 - Potential failure data can be right-censored if it is unknown when the asset will fail in the future.

System Types

- **Impulse Testable** : These systems are maintained in such a way that they are known to be good at the last time they were tested.
 - Assets are often periodically certified.
 - The data is interval censored, from the right and the left.

Module 1.1.3 : Knowing Our Systems

Data is the foundation of reliability assessment

- Reliability analysis requires data describing the system state at known times.
- Data from identical systems tested under similar conditions is preferred.
- When necessary, data from similar systems or operating conditions can be used, if statistical techniques account for the differences.
- SME (Subject Matter Expert) estimates should be used when data are unavailable.

Common reliability data types:

- *Priors*: Historical data or SME estimates used to constrain Bayesian models.
- *Historical*: Data from nearly identical or identical previous systems/components.
- *Time-to-Failure or Repair*: Operating time until failure or repair, including right-censored observations.

Module 1.1.3 : Knowing Our Systems

SYSTEM / Subsystems / Components	HISTORICAL DATA	TIME TO FAILURE DATA	TEST RESULT DATA
Impulse (e.g. Rocket Motor, Bullet)	Aggregated Predictive Models		Reliability Growth Model (Discrete) Observed Model - $S/(S+F)$ or $F/(F+S)$ Single / Multi-Variable Model Non-parametric Model
Testable Impulse (e.g. STANDARD Missile)	Aggregated Predictive Models	Reliability Growth Model (Hybrid)	Reliability Growth Model (Hybrid) Single / Multi-Variable Model Non-parametric Model
Intermittent and Continuous Use (e.g. Radar System, Captive Carry)	Aggregated Predictive Models	System Availability Model Reliability Growth Model Single / Multi-Variable Model Non-parametric Model	Reliability Growth Model (Hybrid) Single / Multi-Variable Model Non-parametric Model

Figure 1: System type versus reliability data type, and notes. Source: “A Simplified Technical Guide for System Reliability: A Reliability Planning and Deployment Reference for Acquisition Professionals.” (Section 219 Project, 2019, Port Hueneme/NAWC China Lake/NSWC Corona).

Module 1.1.3 : Knowing Our Systems

Successful reliability programs require skilled personnel, appropriate data, software tools, and computing resources.

Reliability Modeling

- Skills/Personnel: knowledge of system components, statistics, systems engineering, modeling expertise, software proficiency
- Resources: scored reliability data, modeling software, computing resources

Data Scoring & Evaluation

- Skills/Personnel: system familiarity, to classify valid failures vs. unusable data
- Resources: production/in-service test data, environmental data, reliability databases, computing resources

Reliability Analysis

- Skills/Personnel: engineering, statistics, modeling, and logistics
- Resources: validated reliability models, analysis software, computing resources

Five main questions to ask :

1. What are your program reliability requirements?
2. What is your system type?
3. What types of system data do you have?
4. How much data do you have?
5. In what phase of the life-cycle is your system?

Module 1.1.4 : Reliability Model and Method Selection

Predictive Reliability Models: Aggregated (“Roll-Up”) Models

- Combine subsystem reliability estimates to predict overall system reliability.
- Common methods: MIL-HDBK-217 predictions, Reliability Block Diagrams (RBD)

Example Reliability Roll-Up

Subsystem	Reliability	Running R
Power Supply	0.995	0.995
Processor	0.990	0.985
Communications	0.985	0.970
Sensors	0.980	0.951
System	$R_{\text{sys}} = \prod_i R_i$	0.951

Benefits

- Simple calculations and minimal data requirements
- Can leverage historical data
- Supports reliability allocation during design

Limitations

- Accuracy depends on subsystem models
- Error propagation leads to inaccuracies or imprecision for system performance

Module 1.1.4 : Reliability Model and Method Selection

Reliability Growth Models

- Model improvements in system reliability as failures are identified and corrected during testing.
- Common methods: Crow-AMSAA, Duane Model, and accelerated life testing (HALT/HASS). The example below reflects Crow-AMSAA method, with Cumulative MTBF = Cumulative Hours/Cumulative Failures.

Test Hours	Cumulative Failures	Cumulative MTBF (hr)	Running Growth
100	10	10.0	1.00
300	20	15.0	1.50
700	35	20.0	2.00
1500	50	30.0	3.00

Benefits

- Quantifies reliability improvement
- Measures effectiveness of corrective actions during testing
- Supports reliability planning

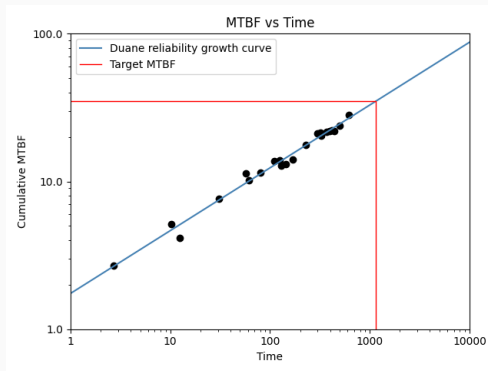
Limitations

- Requires representative data
- Assumes corrective actions are implemented
- Predictions depend on test conditions and reporting

Module 1.1.4 : Reliability Model and Method Selection

Reliability Growth Models

- Model improvements in system reliability from corrective actions during testing.
- Common methods: Crow-AMSAA (continuous and discrete), Duane Model, and accelerated life testing (HALT/HASS/TAAF).



Module 1.1.4 : Reliability Model and Method Selection

System Availability Model

- Model probability that systems are available for operation at an arbitrary time.
- Examples: $A_0 = \text{Uptime} / (\text{Uptime} + \text{Downtime})$, Markov Modeling

Benefits

- Well-defined with low data requirements
- Works for most continuous and impulse testable systems

Limitations

- Not well-defined for impulse untestable systems

Single/Multivariable Model

- Mathematical equations that model reliability versus time and other factors
- Examples: Exponential and Weibull models

Benefits

- Accurate, precise, supports numerical analysis
- Identification of key parameters

Limitations

- Requires larger data sets, knowledge of statistical and computational techniques

Module 1.1.5 : Model Development

Data is the fundamental driving force for reliability analysis.

- Data is often not cheap
- Carefully consider data acquisition
 - Depends on system type (impulse untestable, impulse testable, continuous)

Examples of data items:

- Pass or Fail data (T and E data)
- Failure Reporting and Corrective Action System (FRACAS)
- Hardware specifications, configuration data, inventory tracking
- Material transfer and transport information

Develop the data that is necessary to achieve reliability goals

Module 1.1.5 : Model Development

Data drives model development.

- Some reliability models require time-dependent data
- Some reliability models require pass-fail data
- Data type can evolve over time - reliability models evolve with it

Hardware drives model development.

- Engineering community and subject matter experts (SMEs) can provide historical insight on known components
- System designs map to tools like RBDs, and fault/success trees

Models that provide quantitative point estimates and confidence limits for reliability parameters are preferred

Reliability models should be validated.

- Examples of comparison between model and reality
 - Graph data on the same axes as model prediction
 - Calculate goodness of fit between data and model
- Peer review
 - Portions of the model may be generalized and unclassified
 - These can be compared to other published research
 - New techniques can be published and scrutinized

Module 1.1.5 : Model Development

Example matrix for Reliability Model Selection

SYSTEM / Subsystems / Components	HISTORICAL DATA	TIME TO FAILURE DATA		TEST RESULT DATA	
		NO DATA	1 - 5 6 - 150 150+ TONS OF DATA	NO DATA	1 - 29 30 - 1000 1000+ TONS OF DATA
Impulse (e.g. Rocket Motor, Bullet)	Aggregated Predictive Models			Reliability Growth Model (Discrete)	Observed Model - $S/(S+F)$ or $F/(F+S)$
				Single / Multi-Variable Model	Non-parametric Model
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			Reliability Growth Model	Non-parametric Model	
			Single / Multi-Variable Model		
			Non-parametric Model		